AI-POWERED MULTI-BRAND E-BIKE RECOMMENDATION AND SMART MONITORING SYSTEM

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ABSTRACT

The AI-Powered Multi-Brand E-Bike Recommendation and Smart Monitoring System is an intelligent solution that helps users choose the most suitable electric bicycle while continuously monitoring battery health and vehicle performance. The system uses Artificial Intelligence (AI) to recommend e-bikes based on user preferences such as budget, travel distance, speed, and battery capacity across multiple brands.It also integrates an ESP32 microcontroller, smart Battery Management System (BMS), current and temperature sensors, and an LCD/mobile interface to monitor real-time parameters including battery percentage, voltage, current, temperature, and estimated range. The collected data enables predictive battery health analysis and provides alerts for overheating, low battery, and abnormal operating conditions.This project improves decision-making, enhances battery safety, reduces maintenance costs, and promotes efficient adoption of electric mobility through a smart, user-friendly platform.

Keywords: Artificial Intelligence, E-Bike, ESP32, Smart BMS, Battery Monitoring, Multi-Brand Recommendation.

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